OCR Model Comparison

PaddleOCR, GLM-OCR and Tesseract OCR

This document compares three images: Aadhaar, PAN and voter ID. It shows all nine OCR text outputs, response examples, model differences and deployment effects.

Model	Returns	Runs through
PaddleOCR	Raw text; Aadhaar fields	Python / CPU
GLM-OCR	Raw text and metadata	Local Ollama
Tesseract OCR	Raw text and line confidence	Python / CPU

Image 1: Aadhaar



1. PaddleOCR

Model: PP-OCRv5_server_det + en_PP-OCRv5_mobile_rec. A targeted devanagari_PP-OCRv5_mobile_rec pass reads the Hindi name. Package: PaddleOCR 3.7.0.

Input: apeksh.webp (the image shown above).

```
python main.py apeksh.webp --engine paddleocr --output output\comparison
```

Output: extracted text

```
HR RT TE
UNIQUE IDENTIFICATION AUTHORITY OF INDIA
3TET
Download Date: 05/12/2019
YaT:
D/O/
2HT4RH, 3T 1SR RTE, 2T ST.3H
2i, $I1RGS, 834001
Address:
D/O: Sanjiv Kumar Mishra, gopal ganj / upper
bazar, near shani mandir, upper bazar ranchi,
Ranchi G.P.O., Ranchi, Jharkhand, 834001
7458 6035 2502
WWW
P.O. Box No.1947,
idai gov.in
Rengaluru-560 001
AIRD
HROR
GOVERNMENT OF INDIA
mat
3FT
Apeksha Mishra
30H
/D23/07/2003
/FEMALE
7458 6035 2502
T3TETR
A
46419
Goo0oj pa
Download Date: 05/12/2019
अपेक्षा मिश्रा
```

PaddleOCR: JSON response

This is the complete structured response saved by main.py. Dates use YYYY-MM-DD. Empty strings mean that a value was not extracted.


```
{
  "document_type": "Aadhaar",
  "name": "Apeksha Mishra",
  "name_hindi": "अपेक्षा मिश्रा",
  "date_of_birth": "2003-07-23",
  "gender": "Female",
  "aadhaar_number": "745860352502",
  "virtual_id": "",
  "issue_date": "",
  "download_date": "2019-12-05",
  "address": {
    "care_of": "Sanjiv Kumar Mishra",
    "house_number": "",
    "locality": "Gopal Ganj / Upper Bazar, Near Shani Mandir, Upper Bazar Ranchi",
    "city": "Ranchi",
    "district": "Ranchi",
    "state": "Jharkhand",
    "pin_code": "834001",
    "country": "India"
  },
  "issuing_authority": "UIDAI",
  "qr_code_present": true
}
```


The raw OCR text contains recognition errors. The additional field parser and Hindi-name pass produce the structured fields shown here.

2. GLM-OCR

Model: glm-ocr:q8_0. Runtime: local Ollama. Module: glm_ocr.py.

Input: apeksh.webp (the image shown above).

```
python main.py apeksh.webp --engine glm
```

Output: extracted text

```
आयात
भारतीय विश्वास पहचान प्राधिकरण
UNIQUE IDENTIFICATION AUTHORITY OF INDIA
पता:
D/O: संजीव कमार मिश्रा, गौपालल गौपज / अपर बाजार,
शाखी मठिर के पासस, अपर बाजार राखी, राखी जी.पी.आ
राखी, दुरार्थड, 834001
Address:
D/O: Sanjiv Kumar Mishra, gopal ganj / upper
bazar, near shani mandir, upper bazar ranchi,
Ranchi G.P.O., Ranchi, Jharkhand, 834001
Download Date: 05/12/2019
7458 6035 2502
www
uidai.gov.in
P.O. Box No.1947,
Bengaluru-560 001
भारत सरकार
GOVERNMENT OF INDIA
अपेद्धा मिश्रा
Apeksha Mishra
जन्म तिनिय / DOB: 23/07/2003
मिश्रा / FEMALE
7458 6035 2502
मेरा आयात, मेरी पहचान
```

JSON response: excerpt

The text field is shown in full above. This excerpt shows the model and generation metadata.


```
{
  "engine": "glm",
  "model": "glm-ocr:q8_0",
  "truncated": false,
  "usage": {
    "prompt_tokens": 849,
    "generated_tokens": 445,
    "total_duration_ns": 37862779000
  }
}
```


Observation: the English name, DOB and address were read. Some Hindi words were misread. The current GLM route does not create separate Aadhaar fields.

3. Tesseract OCR

Model: Tesseract 5.4.0 with English language data (eng). Python wrapper: pytesseract 0.3.13.

Input: apeksh.webp (the image shown above).

```
python main.py apeksh.webp --engine tesseract --output output\comparison
```

Output: extracted text

```
= Gediatartrensa wif _
‘BN. UNIQUE)IDENTIFICATION/AUTHORITY OF INDIA
War.
D/O: Satta Hare FAT, HTT TT / HTT ATA
alfa Afex & Oe, rT ah aa ah aH
If, ARGS, 834001
® Address:
& D/O: Sanjiv Ku
bazar, near shani ma
ancni G.P.O., Ranchi, J
@42q peojuMog
mar Mishra, gopal ganj / upper
indir, upper bazar ranchi,
warknand, 634001
SO
R
602/24,
P.0. Box No.1947,
Renaaluru-560 001
Toa Fat
|
. Apeksha Mishra
wem fete / DOB: 23/07/2003 meres
Ar Tert
,| after / FEMALE 5 fe
ae ae eae
i eo
7458 6035 2502
f
\
```

JSON response: excerpt

The full text is shown above. This excerpt includes the first line and its confidence score. The complete saved response contains all line records.


```
{
  "engine": "tesseract",
  "language": "eng",
  "lines": [
    {
      "text": "= Gediatartrensa wif _",
      "confidence": 0.39749999999999996
    }
  ]
}
```


Observation: the name, DOB and Aadhaar number appear in the output. Some address lines are split or misread. English-only language data does not reliably read Hindi.

Image 2: PAN card

Input: pan.png | 600 × 400 pixels. The same image is used for all three models.



All three read D MANIKANDAN, DURAISAMY, 16/07/1986 and BNZPM2501F. Hindi headings and the signature area contain errors or missing text. These are observations from this image, not an accuracy score.

PAN: PaddleOCR response

PaddleOCR

Model: PP-OCRv5_server_det + en_PP-OCRv5_mobile_rec. Raw English OCR; no Aadhaar parser.

Input: pan.png (image shown on this section's first page).

Output: full extracted text

```
3TRC
FEHTST
HRA
HRCOR
INCOME TAX DEPARTMENT
GOVT. OF INDIA
D MANIKANDAN
DURAISAMY
16/07/1986
Permanent Account Number
BNZPM2501F
manifamalo
0008000
Signature
```

Response: JSON excerpt


```
{
  "engine": "paddleocr",
  "language": "en",
  "lines": [
    {
      "text": "3TRC",
      "confidence": 0.4523405432701111
    }
  ]
}
```


The text field is shown above in full. This excerpt shows the first line only; the complete response file includes every line, confidence score and polygon.

PAN: GLM-OCR response

GLM-OCR

Model: glm-ocr:q8_0 through local Ollama.

Input: pan.png (image shown on this section's first page).

Output: full extracted text

```
आयकर विभाग
INCOME TAX DEPARTMENT
D MANIKANDAN
DURAISAMY
16/07/1986
Permanent Account Number
BNZPM2501F
Signature
भारत सरकार
GOVT. OF INDIA
04082011
```

Response: JSON excerpt


```
{
  "engine": "glm",
  "model": "glm-ocr:q8_0",
  "truncated": false,
  "usage": {
    "prompt_tokens": 306,
    "generated_tokens": 94,
    "total_duration_ns": 136897519400
  }
}
```


The text field is shown above in full. Image sizes, text and metadata are in the complete response file.

PAN: Tesseract OCR response

Tesseract OCR

Model: Tesseract 5.4.0; English language data (eng).

Input: pan.png (image shown on this section's first page).

Output: full extracted text

```
steax feat Rd Gee
INCOME TAX DEPARTMENT
D MANIKANDAN
DURAISAMY
16/07/1986
Permanent Account Number
BNZPM2501F
```

Response: JSON excerpt


```
{
  "engine": "tesseract",
  "language": "eng",
  "lines": [
    {
      "text": "steax feat Rd Gee",
      "confidence": 0.14
    }
  ]
}
```


The text field is shown above in full. This excerpt shows the first line only; the complete response file includes every line and confidence score.

Image 3: Voter ID

Input: vote.png | 768 × 954 pixels. The same front-and-back image is used for all three models.



PaddleOCR and GLM read the English name, birth date and voter ID number. Tesseract reads the voter ID number and part of the address, but misses the name and birth date. The Marathi text contains errors in the outputs.

Voter ID: PaddleOCR response

PaddleOCR

Model: PP-OCRv5_server_det + en_PP-OCRv5_mobile_rec. Raw English OCR; no Aadhaar parser.

Input: vote.png (image shown on this section's first page).

Output: full extracted text

```
HRG EasUC 3U
ELECTION COMMISSION OF INDIA
ZSL6279350
U5t91
:44G3
Name: Mamatadevi Agarwal
4aF:44R3
Husband's Name: Ballabhkumar:Agarwal
/Gender/ Femäle
GHGRRE/C4:
Date of Birth / Age: 02-06-1970
e-Electors Photo Identity Card - §-d 3
4:53..230/1,-1423,23
au,,5d, qu,46Rg-411039
an
Address: SR.NO.230/1, VINAYAK
RESIDENCY,DIGHI ROAD,BHOSARI,
BHOSARI,HAVELI, PUNE,MAHARASHTRA-
411039
4SGTGU3CT, 207 -2R
Electoraf Registration Officer, 207 - Bhosari
ZSL6279350
Downfoad Date: 18-04-2024
G
1950
https://ceoelection.maharashtra.gov.in/ceo/
```

Response: JSON excerpt


```
{
  "engine": "paddleocr",
  "language": "en",
  "lines": [
    {
      "text": "HRG EasUC 3U",
      "confidence": 0.4441247880458832
    }
  ]
}
```


The text field is shown above in full. This excerpt shows the first line only; the complete response file includes every line, confidence score and polygon.

Voter ID: GLM-OCR response

GLM-OCR

Model: glm-ocr:q8_0 through local Ollama.

Input: vote.png (image shown on this section's first page).

Output: full extracted text

```
भारत निर्वाहपूर्ण आयोग
ELECTION COMMISSION OF INDIA
ZSL6279350
नाव: ममतादेवी अग्रवाल
Name: Mamatadevi Agarwal
पत्रीव नाव: बल्लमहुमार अग्रवाल
Husband's Name: Ballabhkumar Agarwal
लिना / Gender: रत्य / Female
जनतातीय / वय:
Date of Birth / Age: 02-06-1970
e-Electors Photo Identity Card - इ-मतदेवर फोटो अलक्षणपर
पत्रा: एसआर.ने.230/1, विनायक सेजीडेवी, दियी रोड,
थोसरी, थोसरी, हवेली, पुँप, महाराष्ट्र-411039
Address: SR.NO.230/1, VINAYAK
RESIDENCY,DIGHI ROAD, BHOSARI,
BHOSARI, HAVELI, PUNE, MAHARASHTRA-
411039
मतदेवर जोवदी अलक्षकरी, 207 - थोसरी
Electoral Registration Officer, 207 - Bhosari
Download Date: 18-04-2024
ZSL6279350
1950 https://ceoelection.maharashtra.gov.in/ceo/
```

Response: JSON excerpt


```
{
  "engine": "glm",
  "model": "glm-ocr:q8_0",
  "truncated": false,
  "usage": {
    "prompt_tokens": 930,
    "generated_tokens": 512,
    "total_duration_ns": 40803594300
  }
}
```


The text field is shown above in full. Image sizes, text and metadata are in the complete response file.

Voter ID: Tesseract OCR response

Tesseract OCR

Model: Tesseract 5.4.0; English language data (eng).

Input: vote.png (image shown on this section's first page).

Output: full extracted text

```
sia Prasuper Sirah
& ELECTION COMMISSION OF INDIA
ZSL6279350
| Date of Birth / Age:
e-Electors Photo Identity Card - $-4deR Sic} saa
= Wal: Vasn.4.230/1, arses eehSe ee as,
Sa vite, sited, gach, qt, werre- 411039
Address: SR.NO.230/1, VINAYAK
RESIDENCY, DIGHI ROAD,-BHOSARI,
: ARI; HAVELI, PUNE, MAHARASHTRA-
siftenré, 207 - sient
Registration Officer, 207 - Bhosari
Date: 18-04-2024
ropes :
ZSL6279350
https://ceoelection.maharas! Al
@ Mositecosiecion maheresira. gov
|
|
```

Response: JSON excerpt

```
{
  "engine": "tesseract",
  "language": "eng",
  "lines": [
    {
      "text": "sia Prasuper Sirah",
      "confidence": 0.30666666666666664
    }
  ]
}
```

The text field is shown above in full. This excerpt shows the first line only; the complete response file includes every line and confidence score.

What is different?

These are three different OCR engines. The table describes the versions and settings used in this project.

Feature	PaddleOCR	GLM-OCR	Tesseract OCR
How it reads	Detects text regions, then recognizes each region.	Vision-language model generates text from the image. [2]	OCR engine uses installed language data. [3]
Current setup	PP-OCRv5; Python; CPU. [1]	glm-ocr:q8_0; local Ollama service.	Tesseract 5.4.0 + pytesseract; English (eng).
Saved response	Text, line scores and polygons. Extra parser creates Aadhaar fields.	Text, model, image sizes, truncation flag and token metadata.	Text and line scores. This wrapper does not save boxes.
Hindi / Marathi	English passes misread Indic text. Aadhaar has a targeted Hindi-name pass.	Returned Indic text, but several words are wrong.	These tests use English only. Other languages need traineddata. [3]
Current limitation	Structured parser supports Aadhaar only.	Current route returns text, without separate ID fields.	Current route returns text, without separate ID fields.

A confidence score is not a guarantee. Paddle and Tesseract scores are calculated differently, so do not rank engines by those numbers. The GLM response here has no OCR confidence score.

What did the three images show?

Image	PaddleOCR	GLM-OCR	Tesseract OCR
Aadhaar	Reads English name, DOB and number. Targeted pass reads Hindi name; parser creates fields.	Reads English name, DOB and number. Hindi name is misread.	Reads English name, DOB and number; address is split and noisy.
PAN	Reads English name, DOB and PAN number.	Reads English name, DOB and PAN number.	Reads English name, DOB and PAN number.
Voter ID	Reads name, DOB and voter ID; some labels and Marathi text are wrong.	Reads name, DOB and voter ID; several Marathi words are wrong.	Reads voter ID and part of address; name and DOB are missing.

There is no perfect model in these examples. The Aadhaar image also appears to show different PIN digits in its Hindi and English address lines; review the image before treating either OCR value as verified.

How these results were collected

The Aadhaar results and all GLM responses were saved on 23 October 2026. PAN and voter ID were tested again with PaddleOCR and Tesseract on 26 October 2026. Saved JSON and text files were checked for matching text.

PaddleOCR's Aadhaar route includes extra Hindi-name and date recognition. PAN and voter ID use --raw with English recognition. This is a comparison of the current application settings, not an equal-language model benchmark.

The document shows all nine text outputs without correcting OCR errors. JSON examples are marked as excerpts; the linked response file contains all nine complete saved responses, including all line records and the structured Aadhaar JSON.

Open complete responses (all nine model/image results)

Can we combine all three?

Yes. One application can offer PaddleOCR, GLM-OCR and Tesseract through one upload screen or API. They remain three separate engines; adding them together does not train a new model.

Suggested flow: Upload image → choose document type → run selected OCR → parse fields → validate → return result. If required fields are missing, try another engine or request review.

Option	Benefit	Effect after deployment
Choose one engine	One OCR call for each image.	Only that engine does the work. Simplest way to control resource use.
Use a fallback	Try another engine when required fields are missing or invalid.	Can recover missed fields. A retry adds processing time and compute cost.
Run all three in sequence	Return all outputs for comparison.	Time is roughly the sum of three runs plus overhead. Lower peak memory is possible if each worker releases resources.
Run all three in parallel	Get three outputs at the same time.	Needs more memory and compute. With enough independent resources, time is near the slowest run plus overhead; shared hardware may be slower.

More engines can provide extra evidence, but accuracy does not automatically improve. They can make the same mistake or disagree. Keep each model's original output and flag conflicting ID numbers, names or dates for review.

A practical starting choice

Start with one engine per request. For this project, PaddleOCR is a reasonable first route for Aadhaar because its field parser already exists. Keep GLM as an optional second read for difficult layouts, and Tesseract as an English OCR alternative. Test this routing on more representative images before choosing production defaults.

What changes after deployment?

The following are expected engineering effects, not measurements from a deployed server.

Area	Effect and simple action
Memory	Simultaneous engines and more users increase memory demand. GLM needs memory for model weights, vision processing and context. Limit worker concurrency; measure peak RAM/VRAM. [4]
Response time	First requests may load models. GLM currently uses keep_alive=0, which unloads after a response: lower idle memory, but later calls may reload. Keeping it warm trades memory for response time. [4]
Storage / startup	Package Python dependencies, Paddle weights, the Tesseract binary and language data, and Ollama plus GLM weights. Pre-download models so startup does not depend on downloading them.
Cost	Local inference still uses rented or owned CPU/GPU, RAM, storage and power. Three engines on every image normally do more work than one. No exact monthly cost was measured.
Reliability	One engine can fail or time out. Use a queue, per-engine timeouts and separate workers. Return a partial result with the failed engine named instead of losing successful outputs.
Output format	Normalize engine, model/version, text, fields, errors and elapsed time. Add separate PAN/voter parsers; do not pass these images through the Aadhaar parser.
ID data	Use authenticated uploads and HTTPS. Keep the model service private. Limit retention of images/results and avoid logging full identity numbers.

What is ready, and what is still needed?

Already available in this project

main.py can call each engine. PaddleOCR is the default; GLM uses glm_ocr.py and local Ollama; Tesseract uses its installed executable. Each saves text and JSON. The existing --engine both option runs PaddleOCR and Tesseract only.

Needed before serving users

Add an upload API or web interface, an all-three or fallback router, document-type field parsers, file-size checks, a bounded job queue, timeouts and health checks. These serving features are not implemented by this documentation update.

In separate Docker containers, set --glm-host to the reachable Ollama service address, such as http://ollama:11434. The default 127.0.0.1 points to the application's own container.

Measure several images with one user, then with the expected number of simultaneous users. Record cold/warm response time, peak RAM/VRAM, failure rate and field accuracy against manually checked answers. Use those measurements to choose server size and cost.

Commands used for these examples

```
python main.py apeksh.webp --engine paddleocr --output output\comparison
python main.py apeksh.webp --engine tesseract --output output\comparison
python main.py pan.png --engine both --raw --output output\comparison3
python main.py vote.png --engine paddleocr --raw --output output\comparison3
python main.py vote.png --engine tesseract --output output\comparison3
python main.py apeksh.webp --engine glm
python main.py pan.png --engine glm
python main.py vote.png --engine glm
```

The GLM and Aadhaar commands identify the saved runs from 23 October; they were not rerun for this update. No server was deployed.

Official references

[1] PaddleOCR: PP-OCRv5 models and deployment

[2] GLM-OCR: model and supported deployment options

[3] Tesseract: language data and model choices

[4] Ollama: model unloading, memory and concurrency